

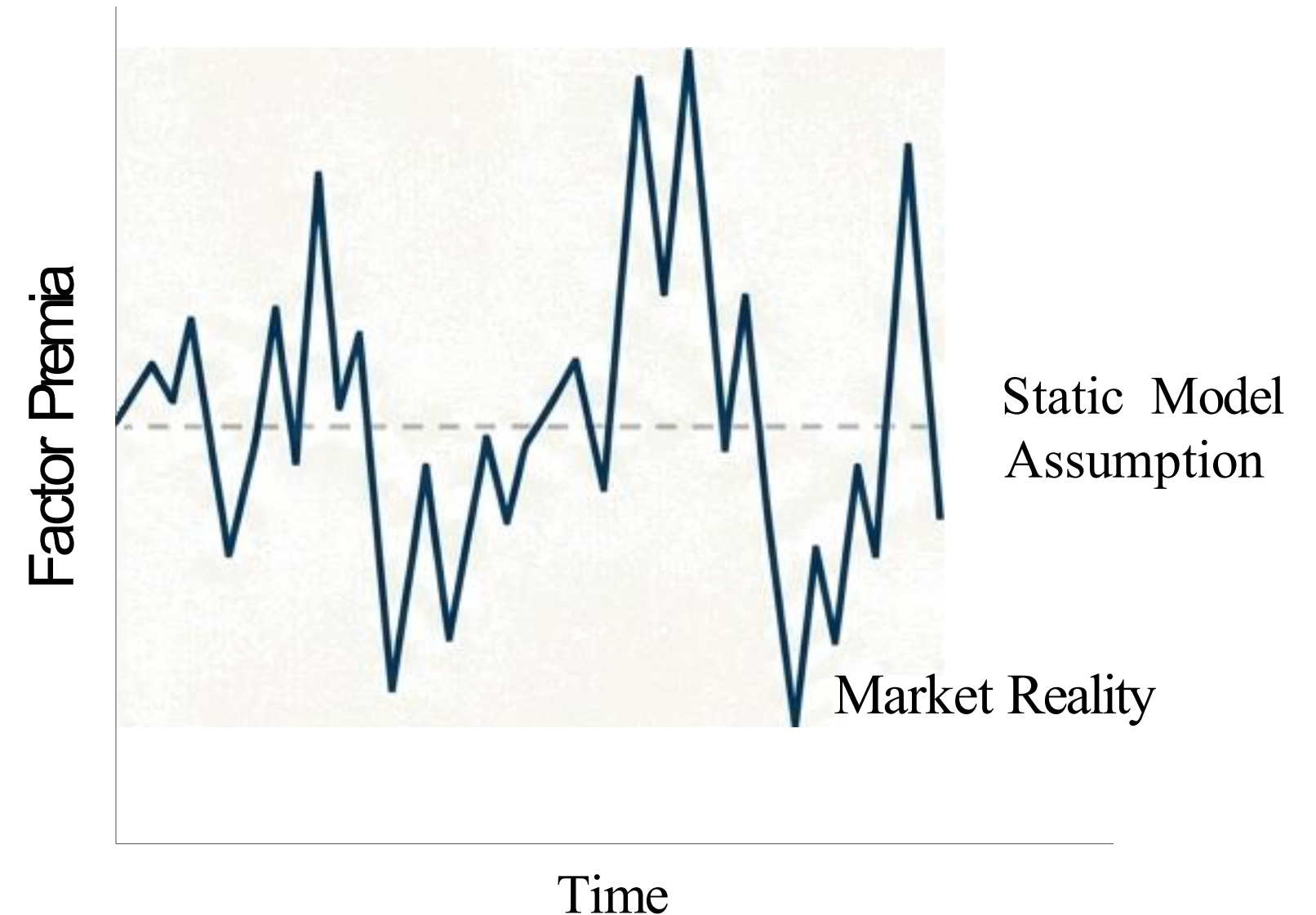
Dynamic Factor Investing with Balanced Temporal Factor Autoencoder

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Motivation: The Limits of Static Factor Models

- Canonical linear models (Fama-French, HXZ) assume linearity and temporal stationarity.
- These assumptions are frequently violated in dynamic markets like China's A-shares, which are characterized by:
 - Frequent regime shifts and structural breaks.
 - Evolving regulatory landscapes.
 - Time-varying factor premia and loadings.
- The Core Problem: Static models fail to capture the nonlinear, time-varying relationships between firm characteristics and future returns, leading to diminished predictive power.



Research Goal and Key Contributions

Overall Goal: To improve cross-sectional return prediction in the CSI 500 universe via a hybrid machine learning pipeline.

- 1. Dynamic Factor Extraction:** Employing rolling Principal Component Analysis (PCA) to create adaptive, denoised features from firm characteristics.
- 2. Sequence Modeling:** Applying a Transformer encoder to capture temporal dependencies within 12-month sequences of factor scores.
- 3. Interpretable Architecture:** Introducing a Temporal Factor Autoencoder (TFA) designed to learn structured, time-varying latent factors.
- 4. Rigorous Evaluation:** Conducting a comprehensive out-of-sample backtest using a rolling-window protocol to assess predictive and economic value.

Data Universe and Characteristics

Universe: CSI 500 constituents, representing medium-sized A-share firms.

Period: 2013—2023 (as per evaluation protocol).

Panel Frequency: Monthly alignment of all variables.

Characteristic Categories:

- Valuation (e.g., P/E, P/B)
- Profitability & Quality
- Leverage & Liquidity
- Growth & Sentiment Measures

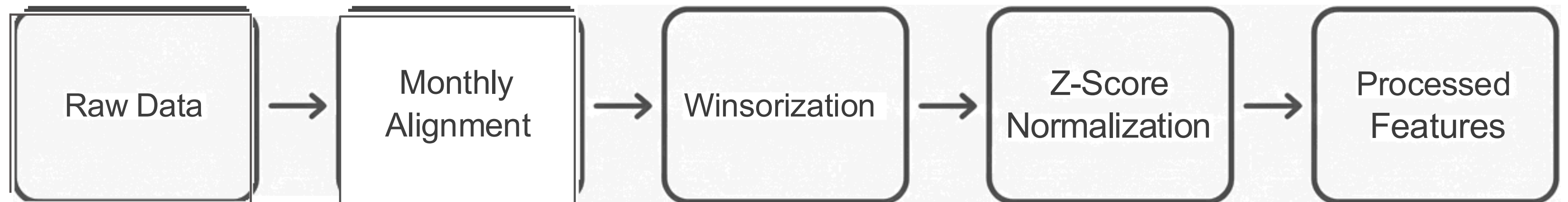
Data Preprocessing Pipeline

- Align accounting and market data on a monthly calendar, forward-filling accounting items.
- Filter out stocks with less than 24 months of historical data.
- For each month, apply cross-sectional transformations:

Winsorization: Clip extreme outliers at the 1st and 99th percentiles.

Normalization: Apply a cross-sectional z-score to standardize feature scales.

Neutralization (Optional): Remove size and industry effects by regressing out exposures.



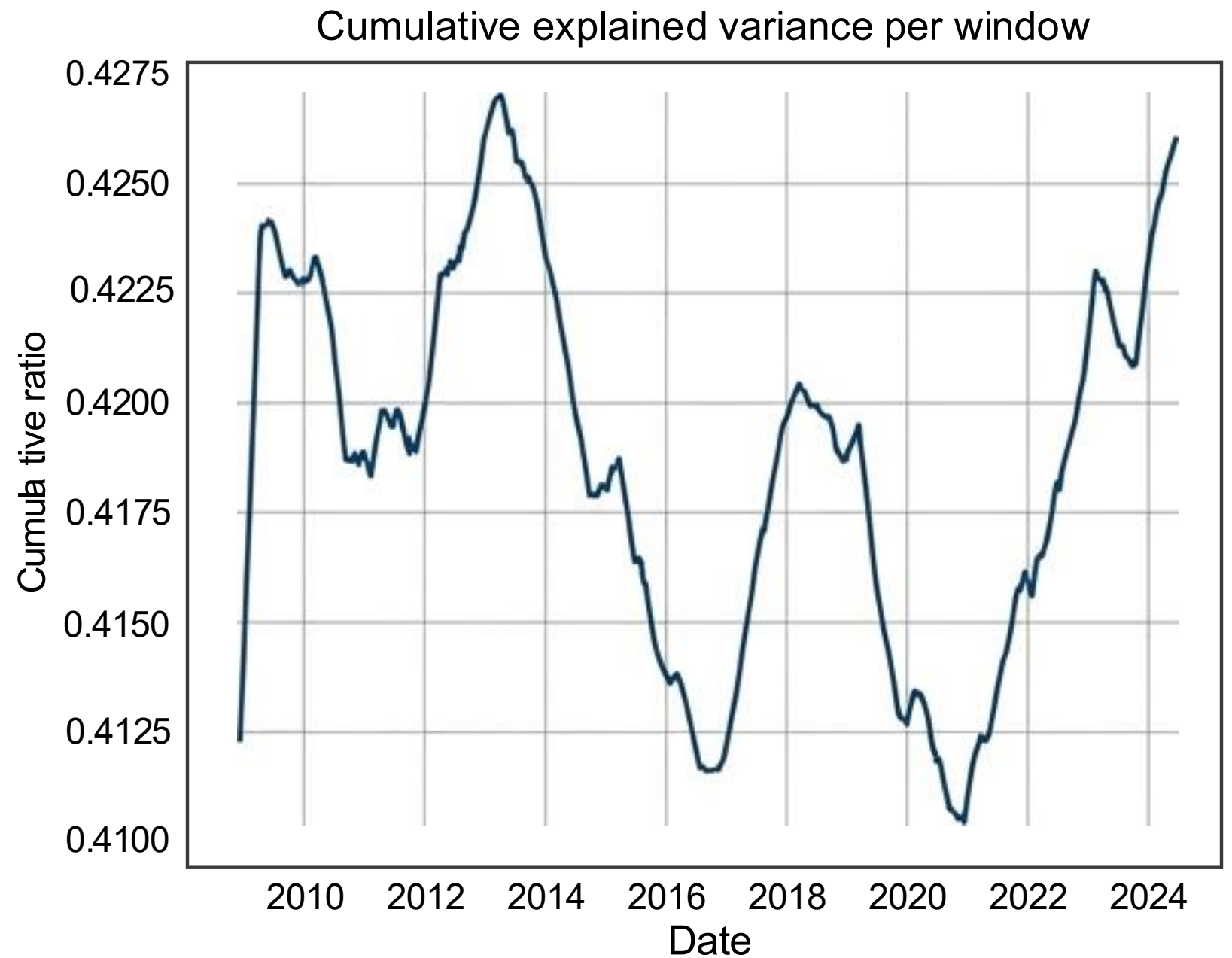
Feature Engineering: Dynamic Factors via Rolling PCA

Mechanism:

- For each rebalancing date t , a 24-month trailing window of processed features is used.
- Drop any feature columns with zero variance within that specific window.
- Component Cap ($N_COMPONENTS = 10$): A hard limit on the maximum number of principal components to extract.
- Variance Target ($VAR_THRESHOLD = 0.80$): A soft target for the cumulative explained variance.

Rationale:

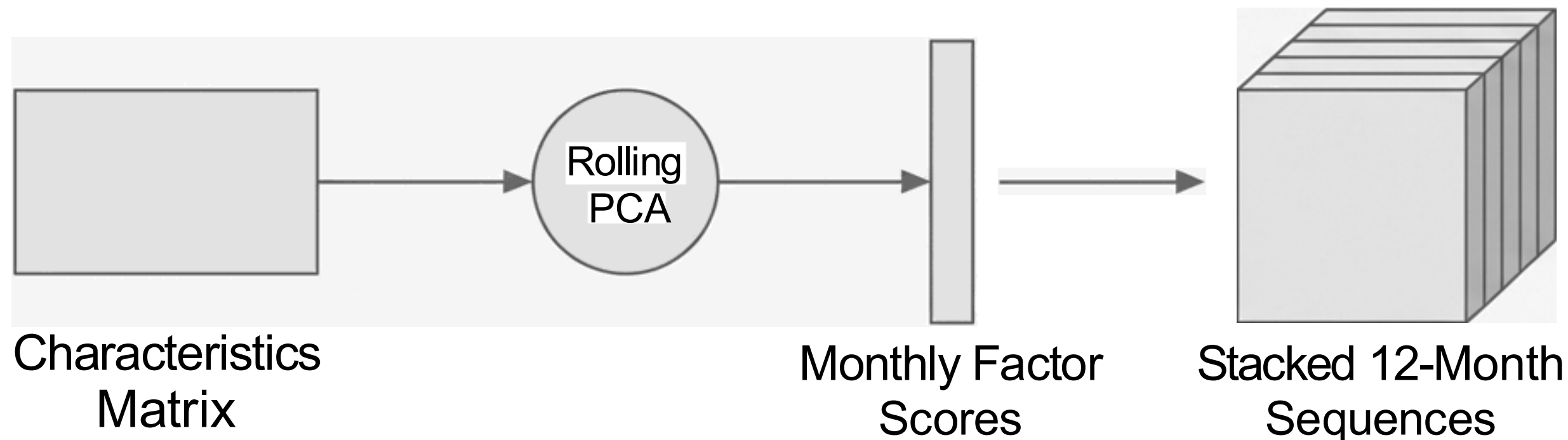
- Reduces dimensionality and denoises collinear characteristics.
- Creates orthogonal factor scores.
- Allows factor loadings to evolve over time, adapting to market conditions.



Input Construction: Factor-Score Sequences

1. The fitted PCA transformation from the rolling window is applied to get factor scores for each stock in month t .
2. For each stock, the most recent 12 months of its factor scores are stacked to form a sequence.
3. Missing data within a sequence is handled via short-term forward-filling (up to 3 months).

Final Input Tensor: A 3D tensor with shape $(N_{\text{stocks}}, T=12, K_{\text{PCA}})$.



Baseline Model Architectures

Common Framework:

- All models are trained on identical rolling windows and input sequences of PCA factor scores.
- Hyperparameters are tuned using nested walk-forward validation within each training window.

Benchmark Suite:

- **Linear:** Ridge Regression (L2 penalty).
- **Nonlinear Ensemble:** Random Forest.
- **Standard Deep Learning:** Multi-Layer Perceptron (MLP).
- **Primary Sequence Baseline:** Transformer Encoder.

Baseline Model Comparison

Model	IC	ICIR	Sharpe	MaxDD
Ridge	0.0212	0.1854	0.8224	0.9953
Transformer	0.0361	0.2706	0.8956	0.6287
MLP	-0.0020	0.2706	-0.0756	1.0073

Core Motivation

Transformer Limitations

- While a powerful predictor, it largely functions as a **'black box'**.
- The learned representations **lack direct economic interpretability**.

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Transformer	0.0361	0.2706	0.8956	0.6287

Core Motivation

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Good Sharpe, but large MaxDD

Motivation for the Temporal Factor Autoencoder (TFA)

Goal of TFA

- To extend the Transformer architecture to explicitly learn **time-varying, interpretable latent factors**.

Desired Factor Properties

- Predictive efficacy for future returns.
- Ability to **faithfully reconstruct** the original factor-score sequences.
- + Temporal stability (smoothness) and low redundancy (orthogonality).

TFA Architecture and Multi-Term Training Objective

Architecture: A temporal autoencoder built on a Transformer backbone.

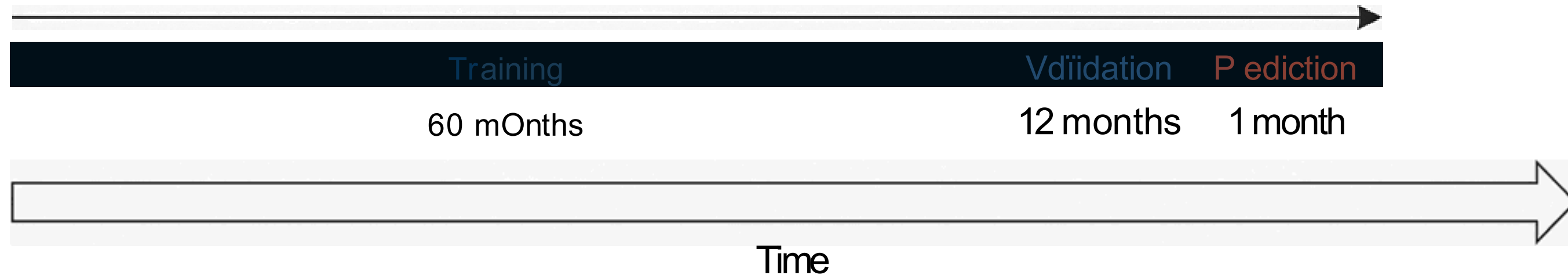
- **Encoder:** A Transformer maps the input sequence to a latent representation.
- **Decoder:** Reconstructs the original PCA factor-score sequence from the latent space.
- **Prediction Head:** Predicts returns from the latent representation.

$$L_{\text{TFA}} = L_{\text{Prediction}} + \alpha \cdot L_{\text{Reconstruction}} + \beta \cdot L_{\text{Smoothness}} + \gamma \cdot L_{\text{Orthogonality}}$$

- **Reconstruction, Smoothness, Orthogonality:** Impose economically desirable structures on the latent factors, ensuring they are meaningful and stable.

Evaluation Protocol

Framework: Rolling-window out-of-sample backtest.



Predictive Metrics

- Information Coefficient (IC)
- IC Information Ratio (ICIR)

Portfolio Metrics

- Long-short decile spread portfolio, rebalanced monthly.
- Metrics calculated after 30 bps trading cost per side: Annualized Sharpe Ratio, Max Drawdown, Turnover.

Main Results: Model Comparison on 11 PCA Factors

Summary Performance Table (Full Backtest Period)

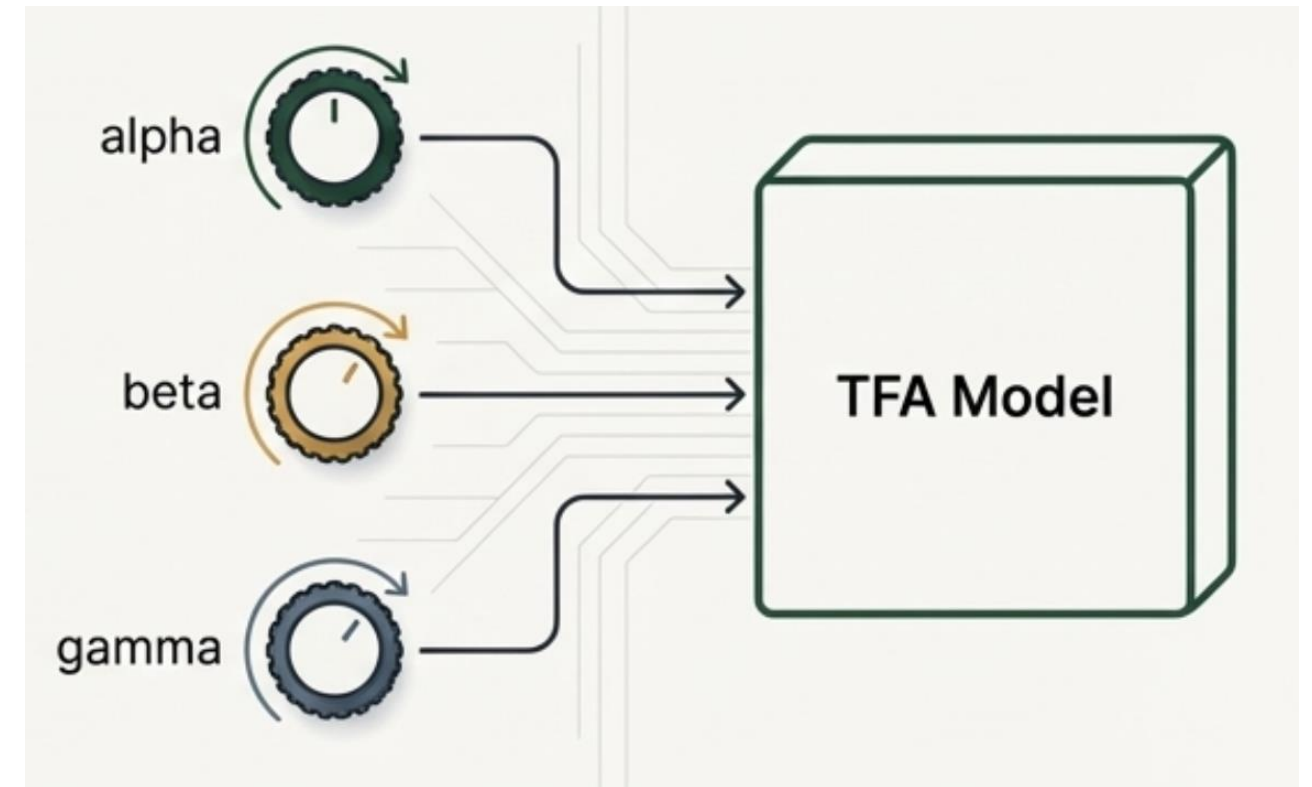
Model	IC Mean	ICIR	Sharpe Ratio	Max Drawdown
Transformer	0.036	0.271	0.896	0.629
TFA	0.022	0.245	0.791	0.479
Ridge	0.021	0.185	0.822	0.995
MLP	-0.002	-0.021	-0.076	1.007

Key Findings:

- * The Transformer model achieves the highest predictive accuracy (IC, ICIR) and Sharpe Ratio.
- * TFA demonstrates competitive performance with a significantly lower maximum drawdown, indicating superior risk control.
- * The standard MLP fails to generalize and performs poorly out-of-sample.

More experiments

Can we improve?



Core Mechanism: Loss function

$$L_{\text{TFA}} = L_{\text{Prediction}} + \alpha \cdot L_{\text{Reconstruction}} + \beta \cdot L_{\text{Smoothness}} + \gamma \cdot L_{\text{Orthogonality}}$$

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- Reconstruction Weight ('alpha'): Governs the importance of reconstructing **input features**. A high 'alpha' can suppress the predictive signal.
- Smoothing/orthogonality ('beta', 'gamma'): Enforce temporal smoothness and factor orthogonality, contributing to **model stability**.

More experiments

Can we improve?

TFA Configuration	IC	ICIR	LS Sharpe	MaxDD	Key Finding
Remove Reconstruction (alpha=0)	0.0352 (↑)	0.3085 (↑)	0.3556 (↓)	0.4518 (↓)	Prediction improves dramatically, but Sharpe collapses.
Remove Smooth/Orthog (beta=0, gamma=0)	0.0094 (↓)	0.0697 (↓)	0.4905 (↓)	0.5384 (↑)	Prediction severely degrades; these components are essential.

Conclusion

- The alpha parameter is the primary lever for managing the prediction-risk trade-off.
- The beta and gamma parameters are crucial for model stability and should not be removed.

More experiments

Our Best Result

Metric	Transformer (Max Prediction)	Optimised TFA (Default)	TFA with `alpha=0.02` (Balanced)
Information Coefficient (IC)	0.0361	0.0218	0.0186
Long-Short Sharpe	0.8956	0.7905	0.8334
Max Drawdown (MaxDD)	0.6287	0.4785	0.2473

More experiments

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Max Drawdown (MaxDD)	0.6287	0.4785	0.2473

Maximum Drawdown that is 50% lower than the standard optimized TFA and 60% lower than the Transformer.

This presents a highly compelling profile for a production strategy focused on capital preservation and consistent returns.

Conclusion and Future Work

Key Takeaways

- A **hybrid pipeline** of rolling PCA and attention-based models is a robust framework for return prediction in the A-share market.
- The Transformer excels at capturing nonlinear, dynamic patterns, leading to superior predictive accuracy.
- TFA successfully **imposes an interpretable economic structure** via its **multi-term loss**, achieving strong performance with improved risk characteristics (lower drawdown).
- Feature engineering is critical; a more parsimonious factor set (11 PCs) yielded better out-of-sample results than a larger one (31 PCs).

Future Work

- Incorporate macroeconomic state variables into the model.
- Extend the TFA framework to **multi-horizon forecasting**.
- Test the methodology on other asset classes or international markets.
- Try more different numbers of PCs as input data to train the TFA framework.

Thanks for your Attention!

<https://github.com/LambertAlpha/Alpha-Hunter>